SILENT ALIASING IN FIXED FOURIER-FEATURE COORDINATE MODELS

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ABSTRACT

Coordinate models built on a *fixed* finite set of Fourier features—the linear core of Fourier-feature implicit neural representations—fit signals by least squares on an exponential dictionary Λ. We give an identifiability theory for what such models do to out-of-band content. (i) On any subset of a rate-Q sampling grid, an integer tone congruent to a model frequency modulo Q is *exactly* indistinguishable from that atom: zero training residual, full-energy reconstruction failure, and—by a two-point argument—no estimator whatsoever can avoid the error (*silent aliasing*). (ii) Random sampling provably destroys this exact aliasing: under timing jitter the tone's visibility follows the jitter's characteristic function ($v \approx 2\pi|r|Q\sigma_t$ for small Gaussian jitter), and under i.i.d. sampling the worst-case aliasability over a finite candidate set concentrates at $O(\sqrt{m \log(mK/\delta)/N})$. (iii) Detection is dichotomous: exactly coherent folds are undetectable by *any* sample-only test (identical observation laws), while visible tones admit exact noncentral-χ^2 detection power, which calibrated sample-only detectors match empirically. Every closed form is validated against Monte Carlo; trained nonlinear networks (Fourier-feature MLPs, SIRENs) are studied strictly as empirical extrapolation, with fold patterns predicted by each network's own NTK linearization and all seed-level failures reported.

***Index Terms*—** Sampling theory, aliasing, identifiability, Fourier features, coordinate networks.

1. INTRODUCTION

Coordinate networks with Fourier features are now standard continuous signal models [1, 2, 3]. Their analytically exact core—and the object of every theorem in this paper—is the *fixed Fourier-feature coordinate model*: a finite frequency set $\Lambda = \{\omega_1, \dots, \omega_m\} \subset \mathbb{R}$ with linear coefficients, fit by least squares. Trained nonlinear networks (Fourier-feature MLPs, SIRENs) enter only as *empirical extrapolation* in Sec. 4, and adaptive or learned frequency sets are outside the scope of our theory; we do not claim results for "all INRs".

Questions. When a signal contains energy outside Λ, three practical questions arise. **Q1**: when is that energy *exactly* invisible in the samples—fitting perfectly while corrupting the reconstruction? **Q2**: how does the sampling design (grids, jitter, i.i.d. draws) change this? **Q3**: when can a practitioner *detect* the problem from the samples alone?

Contributions. (1) An identifiability framework—sample atoms, equivalence classes, *visibility* $v_T(\nu)$, *aliasability* $a_T(\nu)$—in which exact silent aliasing is classified on arbitrary subsets of sampling grids, with an estimator-independent two-point lower bound (Thm. 1). (2) Function-space accounting: a closed-form continuous Gram matrix with Riesz bounds and an exact error decomposition that keeps coefficient error, aliasing-induced modeled-component error, and truncation error *separate* (Thm. 2). (3) Proof that *random sampling breaks exact aliasing*, with explicit rates: a characteristic-function law for jitter and a finite-candidate concentration bound for i.i.d. sampling (Thm. 3). (4) A detection dichotomy: impossibility for coherent folds, exact noncentral-χ^2 power otherwise, and a calibrated empirical operating characteristic with baselines (Thm. 4, Sec. 4).

Relation to prior work (per-item increments; full audit in the repository). Classical modulo-N aliasing [4] is the full-grid special case of Thm. 1, which covers *arbitrary* grid subsets. Generalized sampling [5, 6] supplies the least-squares recast we build on; the equivalence-class, sampling-design, and detection results are not part of it. The generalized aliasing decomposition of eGAD [7] computes an aliasing operator for *given* samples; our $a_T(\nu)$ is a column norm of that operator, and our new content is probabilistic control of it *over the sampling design* (Thm. 3). Basis mismatch [8] concerns tones *near* atoms; silent aliasing is the complementary far regime. The Lomb–Scargle spectral window [9] predicts periodogram leakage; we add the estimator-independent indistinguishability statement and detection theory. The structured-dictionary view of INRs [10] motivates the model class but contains no sampling theory; INR sampling theorems to date [11, 12, 13] are achievability results for realizable signals; an NTK bound from missing frequencies [14] gives an error floor but not the fold structure, invisibility, or detection. Empirical SIREN aliasing has been observed [15]; we supply theory for the linear core and a measured, calibrated diagnostic.

2. SETUP AND BACKGROUND

Signals live on $[0, 1)$; $e_\omega(x) = e^{i2\pi\omega x}$. The model class is $\mathcal{M}_\Lambda = \{\sum_k c_k e_{\omega_k}\}$ with Λ *fixed*. Samples $T = \{t_1, \dots, t_N\} \subset [0, 1)$ define $S_T f = (f(t_j))_j$, the synthesis matrix $\boldsymbol{\Phi} \in \mathbb{C}^{N\times m}$, $\boldsymbol{\Phi}_{jk} = e_{\omega_k}(t_j)$, and observations $\boldsymbol{y} = S_T f + \boldsymbol{\varepsilon}$. We write $f = f_{\rm in} + f_{\rm out}$ with $f_{\rm in} = \sum_k c_k^\star e_{\omega_k}$ and $f_{\rm out} = \sum_l a_l e_{\nu_l}$, $\nu_l \notin \Lambda$. Real signals are fit with the equivalent real cosine/sine design (Hermitian symmetry enforced in the parameterization; see the code). Ordinary least squares $\hat{\boldsymbol{c}} = \boldsymbol{\Phi}^\dagger \boldsymbol{y}$ is used throughout the theory (no ridge); the underdetermined minimum-norm case is treated separately in the code and experiments.

Lemma 1 (LS stability; standard)**.** *If* $\sigma_{\min}(\boldsymbol{\Phi}) > 0$ *and* $f_{\rm out} = 0$ *then* $\|\hat{\boldsymbol{c}} - \boldsymbol{c}^\star\| \le \|\boldsymbol{\varepsilon}\|/\sigma_{\min}(\boldsymbol{\Phi})$ *and, for white noise,* $\mathbb{E}\|\hat{\boldsymbol{c}} - \boldsymbol{c}^\star\|^2 = \sigma^2 \operatorname{tr}((\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1})$.

This is textbook least-squares perturbation, stated for calibration of what follows; the average-case bias/variance decomposition under random out-of-band content is likewise standard bookkeeping and is deferred to the supplement (Prop. S1).

Code, data provenance, and the supplement with full proofs: `github.com/appleweiping/inr-aliasing-limits`.

3. MAIN RESULTS

3.1. T1: exact equivalence and silent aliasing

For $\nu \in \mathbb{R}$ define the *sample atom* $\phi_\nu = (e^{i2\pi\nu t_j})_j$, the *visibility* and *aliasability*

$$v_T(\nu) = \tfrac{1}{\sqrt{N}}\|(\boldsymbol{I} - \boldsymbol{P}_\Lambda)\phi_\nu\|, \qquad a_T(\nu) = \|\boldsymbol{\Phi}^\dagger \phi_\nu\|, \quad (1)$$

with $\boldsymbol{P}_\Lambda$ the orthogonal projector onto range $\boldsymbol{\Phi}$. $v_T(\nu) \in [0,1]$ is the fraction of the tone's sample energy the model cannot absorb; $a_T(\nu)$ is the coefficient bias a unit tone induces.

Theorem 1 (Exact equivalence on grid subsets)**.** *Let $\Lambda \subset \mathbb{Z}$, let T be* any *subset of the rate-Q grid $\{0,\dots,Q-1\}/Q$ with $\sigma_{\min}(\boldsymbol{\Phi}) > 0$ (which requires $N \geq m$ and the elements of Λ distinct mod Q), and let $\nu \in \mathbb{Z}$. (i) $v_T(\nu) = 0$ iff $\phi_\nu \in$ range $\boldsymbol{\Phi}$, i.e. iff a tone at ν is exactly reproducible by the model class on the samples. If $\nu \equiv \omega_k \pmod Q$ then $\phi_\nu = \phi_{\omega_k}$, hence $v_T(\nu) = 0$* for every such subset T*, the fit of a unit tone is the unit coefficient on atom k, and the training residual is exactly zero. (ii) (estimator-independent converse) With $f_1 = f_{\rm in} + a\, e_\nu$ and $f_2 = f_{\rm in} + a\, e_{\omega_k}$, $\nu \equiv \omega_k \pmod Q$, $\nu \neq \omega_k$: $S_T f_1 = S_T f_2$, so under any noise law the two hypotheses induce identical observation distributions, and* any *estimator $\hat f$ satisfies* $\max_{i\in\{1,2\}}\|\hat f - f_i\|_{L^2[0,1)} \geq |a|/\sqrt{2}$. *(iii) If $\nu \not\equiv \omega_k \pmod Q$ for all k and $[\boldsymbol{\Phi}\ \phi_\nu]$ has full column rank, then $v_T(\nu) > 0$.*

Sketch. (i) $e^{i2\pi\nu g/Q}$ depends on ν mod Q; (ii) is a two-point (Le Cam) argument with $\|e_\nu - e_{\omega_k}\|_{L^2} = \sqrt{2}$; (iii) is the rank condition. Full proofs in the supplement. *Silent aliasing* is regime (i)–(ii): the model fits the samples to the noise floor while the reconstruction error is at least the full out-of-band energy, at any budget N satisfying the rank condition, with ν arbitrarily far from Λ. Classical aliasing is the special case T = full grid ($Q = N$) [4]; the point here is that *nonuniform designs inherit the grid's equivalence classes* whenever they live on a grid.

3.2. T2: function-space accounting

Coefficient norms are not signal energies unless the dictionary is orthonormal. Let $G_{L^2}[k,\ell] = \langle e_{\omega_\ell}, e_{\omega_k}\rangle_{L^2[0,1)} = e^{i\pi\delta}\,\mathrm{sinc}(\delta)$, $\delta = \omega_\ell - \omega_k$, $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$.

Theorem 2 (Riesz conversion and exact decomposition)**.** *(i) With $\lambda_{\min}, \lambda_{\max}$ the extreme eigenvalues of G_{L^2} (positive iff the frequencies are distinct), $\lambda_{\min}\|\boldsymbol{c}\|^2 \leq \|\sum_k c_k e_{\omega_k}\|^2_{L^2} \leq \lambda_{\max}\|\boldsymbol{c}\|^2$. (ii) With $\Delta\boldsymbol{c} = \hat{\boldsymbol{c}} - \boldsymbol{c}^\star$,*

$$\|\hat f - f\|^2_{L^2} = \Delta\boldsymbol{c}^* G_{\Lambda\Lambda}\Delta\boldsymbol{c} - 2\,\mathrm{Re}\,\Delta\boldsymbol{c}^* G_{\Lambda,\mathrm{out}}\,\boldsymbol{a} + \boldsymbol{a}^* G_{\mathrm{out}}\,\boldsymbol{a}, \quad (2)$$

the three terms being the aliasing-induced modeled-component *error, a cross term (zero for all-integer frequencies), and the* truncation *energy.*

Equation (2) is elementary but load-bearing: truncation error exists for *any* sampling and is not aliasing; only the first term is the model *misplacing* energy, and only that term responds to sampling design (Fig. 1d).

3.3. T3: random sampling breaks exact aliasing

Theorem 3 (Sampling design)**.** *Let $\Lambda \subset \mathbb{Z}$ with $|\Lambda| = m$. (a)* (jitter) *Let $t_j = g_j/Q + \eta_j$ with g_j grid points and η_j i.i.d., mean-zero, with characteristic function χ_η, and let $\nu = \omega_k + rQ$, $r \in \mathbb{Z}\setminus\{0\}$. Then the empirical fold coherence $\frac{1}{N}\sum_j e^{i2\pi(\nu-\omega_k)t_j}$ has mean $\chi_\eta(2\pi rQ)$ and concentrates around it at rate $2\sqrt{\log(4/\delta)/N}$; consequently $v_T(\nu)^2 = 1 - |\chi_\eta(2\pi rQ)|^2 + O_P\big(m\sqrt{\log(m/\delta)/N}\big)$. For Gaussian jitter of std σ_t, $\chi_\eta(2\pi rQ) = e^{-2(\pi rQ\sigma_t)^2}$, giving the small-jitter law $v_T(\nu) \approx 2\pi|r|Q\sigma_t$. (b)* (i.i.d. samples) *Let t_j be i.i.d. from a density whose limiting normalized Gram satisfies $G_\infty \succeq \lambda_{\min}\boldsymbol{I}$ and whose limiting model–candidate coherences vanish on a finite candidate set Ω, $|\Omega| = K$ (for the uniform density and integer-disjoint candidates, $G_\infty = \boldsymbol{I}$, $\lambda_{\min} = 1$). With $\varepsilon = 2\sqrt{\log\big(4(mK+m^2)/\delta\big)/N}$ and $m\varepsilon < \lambda_{\min}$, with probability at least $1-\delta$,*

$$\max_{\nu\in\Omega}\ a_T(\nu)\ \leq\ \frac{\sqrt{m}\,\varepsilon}{\lambda_{\min} - m\varepsilon}. \quad (3)$$

Sketch. (a) On the grid part the fold phase is exactly 1; the jitter contributes i.i.d. unit-modulus factors, so the coherence is an empirical characteristic function; Hoeffding on real and imaginary parts. (b) Hoeffding + union bound over all $mK + m^2$ coherences, then a perturbation bound on $a_T(\nu) = \|(\boldsymbol{\Phi}^*\boldsymbol{\Phi}/N)^{-1}(\boldsymbol{\Phi}^*\phi_\nu/N)\|$. Proofs in the supplement; both laws are verified to tight tolerance in Fig. 1(a,b).

Three honest distinctions follow. *Exact* grid equivalence persists at every N *if the design stays on the grid* (Thm. 1); for a *fixed* separated tone, randomized designs make aliasing vanish at explicit rates (Thm. 3); and at any finite N an adversary may still pick near-invisible alternatives—but under i.i.d. sampling even the adversarial candidate within a finite set becomes visible (Fig. 1c). We make no claim that "no sampling density removes aliasing"; what no design removes is the truncation term of (2).

3.4. T4: detection—impossibility and power

Theorem 4 (Detection dichotomy)**.** *(a) If $S_T f_1 = S_T f_2$, then for any noise distribution the observation laws under the two hypotheses coincide, and every sample-based test has power equal to its size. (b) For $\boldsymbol{y} = \boldsymbol{D}\boldsymbol{\beta} + \boldsymbol{s} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2\boldsymbol{I})$, $\boldsymbol{D}$ the real design of rank p and $\boldsymbol{s}$ the sample vector of an additive component, the residual statistic $\|(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{y}\|^2/\sigma^2$ is χ^2_{N-p} under H0 and noncentral $\chi^2_{N-p}(\lambda)$ with $\lambda = \|(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{s}\|^2/\sigma^2$ under H1: detection power is driven by* amplitude times visibility*, and is exactly the size α when $\lambda = 0$ (coherent fold).*

Sample-only detectors must be *calibrated*: in all experiments, thresholds come from an independent null calibration set and are then frozen; test-set FPR/TPR are reported separately (Sec. 4).

4. EXPERIMENTS

All identities and bounds above are validated against Monte Carlo in the released test suite (34 tests); experiment JSONs carry seeds, configs, and environment stamps.

Theorem validation (Fig. 1). (a) the jitter law: empirical visibility of a grid-coherent tone matches $\sqrt{1-\chi_\eta^2}$ across two decades, with the exact fold ($v \approx 10^{-13}$) at $\sigma_t = 0$; (b) worst-case aliasability over a K=31 candidate set under i.i.d. sampling decays as $N^{-1/2}$ beneath the proved bound (loose, as unions are, but finite exactly where the theorem says); (c) persistence vs randomization: the grid-subset coherent tone stays exactly invisible at every N, while under i.i.d. sampling both the same tone and the per-N *adversarial* candidate are immediately visible; (d) the decomposition:

jitter drives the aliasing-bias term to the noise level while the truncation floor is untouched —sampling design removes aliasing, never truncation.

Detection (Fig. 2). With calibration/test separation, sample-only detectors (ring test, residual, Lomb–Scargle band test, held-out prediction, cross-fit) track the exact Thm. 4(b) power curve; at the calibrated threshold the test-set FPR stays at its nominal 5%. On the exactly coherent grid fold every detector collapses (ring AUC 0.21, TPR at 5% FPR 0.03)—the indistinguishability of Thm. 4(a) made visible, reported separately from the operating characteristics by design.

Real signals, sample-only. On real series (Mauna Loa CO_2, sunspots, and nine 2-s speech segments from three Open Speech Repository recordings; anti-aliased `resample_poly` decimation; full records, no window selection), bandwidth selectors that see *only the training samples* are compared against an oracle that reads the reference spectrum. On CO_2 the sample-only Lomb–Scargle selector attains RMSE 0.167 vs the oracle heuristic's 0.204 (mean over 10 draws; paired CIs in the released results)—deployable selection is not the bottleneck; cross-validated selection is markedly less stable under gappy sampling. Full per-method, per-signal tables with paired 95% CIs are in the repository.

Trained networks (empirical extrapolation only). For FF-MLPs, SIRENs, and an explicitly band-limited head, tone attribution is measured by an ablation (identical initialization trained with/without the tone) and compared to the prediction of *each network's own NTK linearization*—no arbitrary reference band. Across 20 sampling seeds per condition we report exact-match and top-3 fold agreement rates, pattern-correlation distributions with bootstrap CIs, and controls (label permutation, amplitude, weight-seed stability); seed-level disagreements are reported, not hidden (Fig. 3, left: a representative run; right: correlation distributions). In 2-D (Fig. 4), the linear model reproduces the predicted frequency-vector fold exactly ((a): planted $(27, 3)$ on a rate-32 grid subset folds to $(-5, 3)$, residual 10^{-15}), and trained 2-D networks show the predicted DFT replicas under *lattice* (coherent) masks but not under random masks of equal density—with ridge/early-stopping baselines and clean/noisy/held-out/full-field PSNR reported separately in the released results.

5. LIMITATIONS AND CONCLUSION

Our theorems cover fixed-feature, linear-coefficient Fourier models; trained nonlinear networks are studied only empirically, and adaptive/learned frequency sets are open. The concentration bound is a union bound (loose constants); the jitter law's multi-atom correction is $O_P(m\sqrt{\log/N})$. Within that scope the theory is complete and mechanically verified: exact silent aliasing is a property of *grid designs* and is classified by congruence; randomized sampling provably destroys it at explicit rates; detectability is dichotomous and quantitatively matched by calibrated sample-only detectors. Practical guidance: avoid pure grid designs when out-of-band content is possible; jitter of std σ_t buys visibility $\approx 2\pi|r|Q\sigma_t$; calibrate diagnostics on explicit nulls; and never read training residual as reconstruction fidelity.

6. REFERENCES

[1] V. Sitzmann, J. Martel, A. Bergman, D. Lindell, and G. Wetzstein, "Implicit neural representations with periodic activation functions," in *NeurIPS*, 2020.

[2] M. Tancik, P. Srinivasan, B. Mildenhall, et al., "Fourier features let networks learn high frequency functions in low dimensional domains," in *NeurIPS*, 2020.

[3] B. Mildenhall, P. Srinivasan, M. Tancik, et al., "NeRF: Representing scenes as neural radiance fields for view synthesis," in *ECCV*, 2020.

[4] C. E. Shannon, "Communication in the presence of noise," *Proc. IRE*, vol. 37, no. 1, pp. 10–21, 1949.

[5] B. Adcock and A. C. Hansen, "Generalized sampling and infinite-dimensional compressed sensing," *Found. Comput. Math.*, vol. 16, pp. 1263–1323, 2016.

[6] M. Unser, "Sampling—50 years after Shannon," *Proc. IEEE*, vol. 88, no. 4, pp. 569–587, 2000.

[7] M. K. Transtrum, G. Hart, T. J. Jarvis, and J. P. Whitehead, "eGAD! Double descent is explained by generalized aliasing decomposition," *arXiv:2408.08294*, 2024.

[8] Y. Chi, L. L. Scharf, A. Pezeshki, and A. R. Calderbank, "Sensitivity to basis mismatch in compressed sensing," *IEEE Trans. Signal Process.*, vol. 59, no. 5, pp. 2182–2195, 2011.

[9] J. T. VanderPlas, "Understanding the Lomb–Scargle periodogram," *Astrophys. J. Suppl. Ser.*, vol. 236, no. 1, 2018.

[10] G. Yüce, G. Ortiz-Jiménez, B. Besbinar, and P. Frossard, "A structured dictionary perspective on implicit neural representations," in *CVPR*, 2022.

[11] M. Najaf and G. Ongie, "Towards a sampling theory for implicit neural representations," in *Asilomar Conf. Signals, Syst., Comput.*, 2024, arXiv:2405.18410.

[12] M. Najaf and G. Ongie, "Sampling theory for super-resolution with implicit neural representations," *SIAM J. Imaging Sci.*, 2025, arXiv:2506.09949.

[13] H. Saratchandran, S. Ramasinghe, V. Shevchenko, A. Long, and S. Lucey, "A sampling theory perspective on activations for implicit neural representations," in *ICML*, 2024, arXiv:2402.05427.

[14] K. Ma, M. Zhu, Y. Zhan, H. Yin, S. Wang, and Y. Zheng, "Robustifying Fourier features embeddings for implicit neural representations," *arXiv:2502.05482*, 2025.

[15] H. Chandravamsi et al., "Spectral bottleneck in sinusoidal representation networks: Noise is all you need," *arXiv:2509.09719*, 2025.

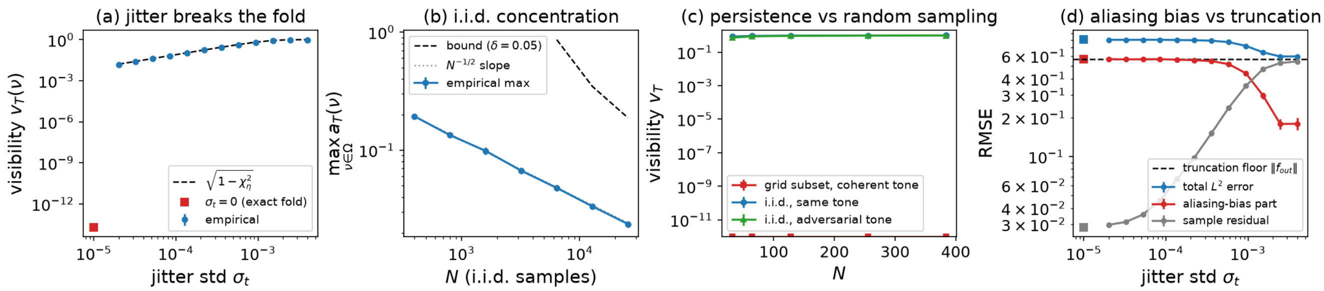

Fig. 1. Theorem validation (mean $\pm$ 95% CI over 20 draws). (a) T3a jitter law; (b) T3b concentration with the proved bound; (c) T1 grid persistence vs i.i.d. sampling (fixed and adversarial tones); (d) T2 decomposition: jitter removes aliasing bias, not truncation.

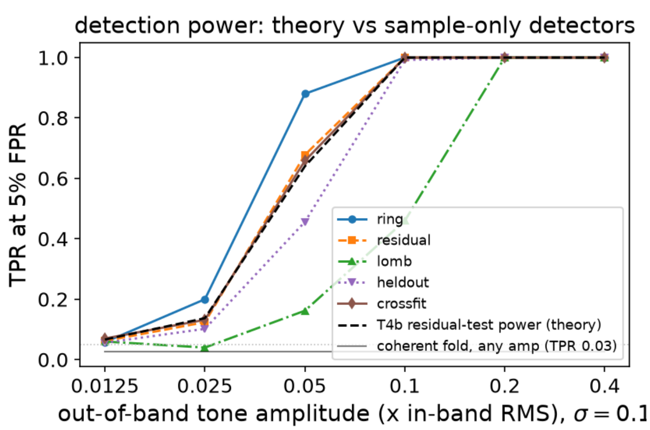

Fig. 2. Detection power (TPR at 5% FPR, calibration/test separated) vs tone amplitude at σ=0.1: five sample-only detectors against the exact Thm. 4(b) power curve; the coherent fold stays at chance at any amplitude (Thm. 4(a)).

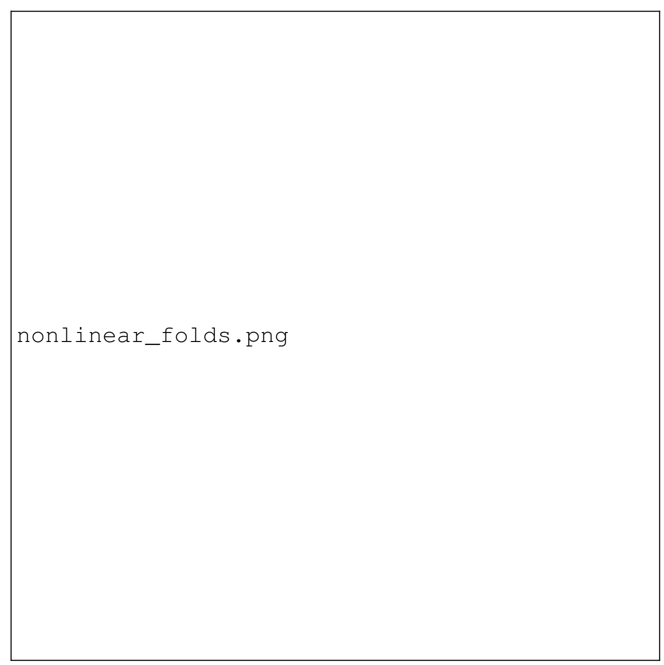

Fig. 3. Trained networks (empirical): ablation-measured tone attribution vs each network's NTK prediction; right: pattern-correlation distributions over 20 sampling seeds (whiskers 5–95%), with the label-permutation control.

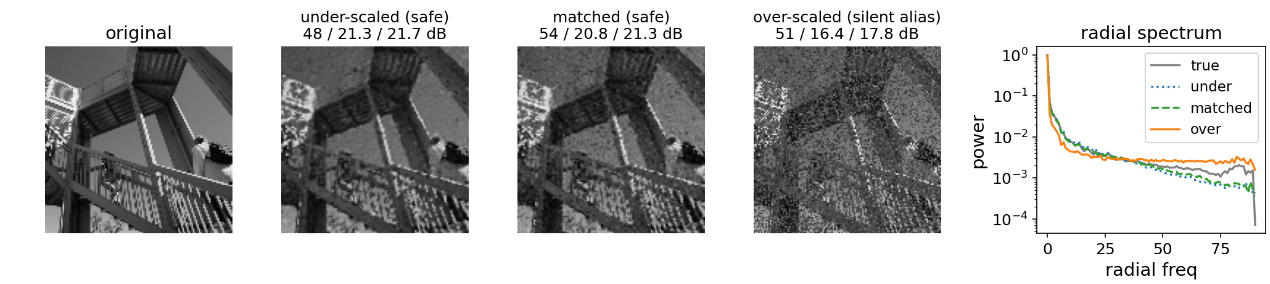

Fig. 4. 2-D structured aliasing. (a) linear model: the planted out-of-band frequency vector folds exactly onto the predicted atom (circle) on a grid subset; (b,c) trained 2-D networks: DFT of over-scaled reconstructions shows the predicted replicas (circles) under a lattice (coherent) mask but not under a random mask of equal density.